

A Standard-Gaussian Counterexample to Equality of the Kim–Milman and Brenier Transport Maps

Automated mathematics run `fa_banach.001`, lane 13
Model: GPT5.6

22 July 2026

Status. Candidate counterexample, likely valid, awaiting human review. The proof is analytic; the included symbolic computation only checks its elementary matrix and integral arithmetic.

1 Source question

Kim and Milman constructed a contraction from a Gaussian measure to a more log-concave measure by reversing an Ornstein–Uhlenbeck heat-flow interpolation [1]. Their Section 6 asked whether this map must coincide with the Brenier optimal-transport map. Tanana proved that the answer is no for a non-standard Gaussian source [2]. The final section of her note observed that the two maps do coincide when both the source is standard Gaussian and the target is Gaussian, but supplied numerical evidence that they need not coincide for a general log-concave perturbation of the standard Gaussian. The relevant page-5 passage is reproduced below.

The question addressed here is therefore:

If the source is the standard Gaussian measure and $d\nu = Z^{-1}e^{-F}d\gamma$ with F convex, must the Kim–Milman map equal the Brenier map?

2 Definitions

Let $\gamma = \gamma_2$ be standard Gaussian measure on $\mathbb{R}^2$, and let $(P_t)_{t \geq 0}$ be its Ornstein–Uhlenbeck semigroup:

$$P_t g(x) = \int_{\mathbb{R}^2} g(e^{-t}x + \sqrt{1 - e^{-2t}}y) d\gamma(y).$$

Given a smooth convex globally Lipschitz F and $\varepsilon > 0$, put

$$d\nu_\varepsilon = Z_\varepsilon^{-1}e^{-\varepsilon F} d\gamma.$$

Following Kim–Milman, define

$$Z_t^\varepsilon = -\log P_t(e^{-\varepsilon F}), \quad \frac{d}{dt}S_t^\varepsilon(x) = \nabla Z_t^\varepsilon(S_t^\varepsilon(x)), \quad S_0^\varepsilon = \text{Id}.$$

Then S_t^ε transports ν_ε to the interpolating measure and S_∞^ε transports ν_ε to γ . The forward Kim–Milman map is $T_{KM}^\varepsilon = (S_\infty^\varepsilon)^{-1}$.

The case of standard normal distribution μ

When μ is the standard normal distribution ($A = \text{Id}$) and ν is Gaussian, the Kim-Milman construction does give the Brenier map ([3, Section 6]). However, our numerical result suggests that this does not hold for general $d\nu = \exp(-F)d\mu$ with convex function F . We consider the case $n = 2$, $F(x) = F(x_1, x_2) = x_1^4 + x_2^4 + (x_1 + x_2)^2$ and the starting point $x = (-0.5, 0)$. Our numerical solution of (3) yielded $S_\infty(x) \approx (-1.054, -0.231)$, while the numerical solution to (4) converged as $t \rightarrow \infty$ to a non-symmetric matrix approximately equal to

$$\begin{pmatrix} 2.303 & 0.441 \\ 0.467 & 2.013 \end{pmatrix},$$

meaning that S_∞ is unlikely to be a gradient of a convex function at point x . To obtain this approximations, we used the explicit Euler method for both ODE's with terminal time $T = 30$ and the following time step sizes:

time interval	Δt
[0,0.1]	0.00002
[0.1,0.5]	0.00005
[0.5,1]	0.0002
[1,3]	0.0005
[1,5]	0.002
[5,30]	0.005

To approximate the semigroup P_t^A , which becomes the Ornstein-Uhlenbeck semigroup when

Figure 1: Tanana's standard-Gaussian question and numerical evidence [2, p. 5].

3 Proof intuition

At first order in ε , the derivative of the inverse flow is an integral of Hessians and is necessarily symmetric. At second order, most terms are still Hessians, but chronological multiplication remembers the order in which different Hessians act. Its skew part is exactly a double integral of commutators

$$\int_{0 < s < t < \infty} [D^2 P_t F(x), D^2 P_s F(x)] ds dt.$$

Thus it suffices to make two OU-smoothed Hessians fail to commute in an integrated sense. A sum of fourth powers of two nonorthogonal linear forms does so explicitly. We first calculate with that polynomial, then replace the scalar fourth power by a smooth convex linear-tail truncation. The commutator coefficient varies continuously under this replacement, whereas global Lipschitz regularity makes the Kim-Milman flow unproblematic. A nonsymmetric inverse-flow derivative is impossible for the inverse of a Brenier map.

4 The perturbative skew formula

Lemma 1 (Well-posedness and integrable parameter expansion). *Let $F \in C^\infty(\mathbb{R}^2)$ be convex, bounded below, and globally Lipschitz, with bounded derivatives of orders at least two. For all sufficiently small $\varepsilon \geq 0$, the flow S_t^ε and its spatial derivative are defined for all $t \geq 0$, converge as $t \rightarrow \infty$, and are twice differentiable in ε at 0. On every fixed compact set, all remainders needed below have an integrable-in- t envelope.*

Proof. The OU commutation identity gives

$$\nabla P_t g = e^{-t} P_t(\nabla g), \quad D^k P_t g = e^{-kt} P_t(D^k g).$$

For $g = e^{-\varepsilon F}$,

$$|\nabla Z_t^\varepsilon| = \frac{|\nabla P_t g|}{P_t g} \leq \varepsilon e^{-t} \|\nabla F\|_\infty.$$

The quotient rule, the same commutation identities, and boundedness of the derivatives of F give, locally uniformly in space and uniformly for small ε ,

$$\|D^k Z_t^\varepsilon\| \leq C_k \varepsilon e^{-kt} \quad (k = 2, 3),$$

with harmless additional products having faster decay. The denominators are uniformly separated from zero on compact sets: after adding a constant assume $F \geq 0$, and use Jensen's inequality and the linear-growth bound on F to bound $P_t(e^{-\varepsilon F})$ below.

The velocity is integrable in time, while its derivative has an integrable bound. Standard ODE estimates therefore give global existence at every fixed initial point, convergence of S_t^ε and DS_t^ε , and uniform compact-set bounds. Differentiating $-\log P_t(e^{-\varepsilon F})$ in ε is justified by Gaussian dominated convergence (a Lipschitz function has only linear growth). The same integrable estimates permit twice differentiating the ODE and passing to $t = \infty$. This proves all assertions. $\square$

Lemma 2 (Second-order skew formula). *Under the hypotheses of Lemma 1, for every $x \in \mathbb{R}^2$,*

$$DS_\infty^\varepsilon(x) - DS_\infty^\varepsilon(x)^T = \varepsilon^2 \mathcal{C}_F(x) + o(\varepsilon^2), \quad (1)$$

where

$$\mathcal{C}_F(x) = \int_0^\infty \int_0^t [D^2 P_t F(x), D^2 P_s F(x)] ds dt. \quad (2)$$

Proof. Write $H_t = D^2 P_t F(x)$. Since

$$-\log P_t(e^{-\varepsilon F}) = \varepsilon P_t F + O(\varepsilon^2),$$

we have, along the perturbed trajectory,

$$D^2 Z_t^\varepsilon(S_t^\varepsilon(x)) = \varepsilon H_t + \varepsilon^2 K_t + o(\varepsilon^2).$$

Here K_t is symmetric: it is the sum of the Hessian of the direct second ε -variation of Z_t^ε and the directional derivative of the Hessian $D^2 P_t F$ in the first variation of the trajectory. The latter remains symmetric in its two matrix indices because third derivatives of a scalar function are symmetric.

Let $J_t^\varepsilon = DS_t^\varepsilon(x)$ and write

$$J_t^\varepsilon = I + \varepsilon A_t + \varepsilon^2 B_t + o(\varepsilon^2).$$

The variational equation

$$(J_t^\varepsilon)' = D^2 Z_t^\varepsilon(S_t^\varepsilon(x)) J_t^\varepsilon$$

gives

$$A_t = \int_0^t H_s ds, \quad B_t' = K_t + H_t A_t.$$

Both K_t and A_t are symmetric. Consequently

$$B_\infty - B_\infty^T = \int_0^\infty (H_t A_t - A_t H_t) dt = \int_0^\infty \int_0^t [H_t, H_s] ds dt.$$

Lemma 1 justifies the integrations and the limiting expansion, which proves (1). $\square$

5 An exact nonzero coefficient

Let

$$a = \begin{pmatrix} 1 \\ 0 \end{pmatrix}, \quad b = \begin{pmatrix} 1 \\ 1 \end{pmatrix}, \quad F_0(x) = (a \cdot x)^4 + (b \cdot x)^4, \quad x_0 = \begin{pmatrix} 0 \\ 1 \end{pmatrix}.$$

Set

$$P = \sum_{v \in \{a, b\}} (v \cdot x_0)^2 v v^T = b b^T = \begin{pmatrix} 1 & 1 \\ 1 & 1 \end{pmatrix},$$

and

$$Q = \sum_{v \in \{a, b\}} |v|^2 v v^T = a a^T + 2 b b^T = \begin{pmatrix} 3 & 2 \\ 2 & 2 \end{pmatrix}.$$

Lemma 3. *For the polynomial model above,*

$$\mathcal{C}_{F_0}(x_0) = \begin{pmatrix} 0 & 6 \\ -6 & 0 \end{pmatrix} \neq 0.$$

Proof. Put $r_t = e^{-2t}$. The Mehler formula and the fourth Gaussian moment give, for any fixed vector v ,

$$D^2 P_t(v \cdot x)^4 = 12 (r_t^2 (v \cdot x)^2 + r_t(1 - r_t)|v|^2) v v^T.$$

Therefore

$$H_t := D^2 P_t F_0(x_0) = 12 (r_t^2 P + r_t(1 - r_t)Q).$$

It follows that

$$[H_t, H_s] = 144 r_t r_s (r_t - r_s) [P, Q].$$

Direct integration yields

$$\int_0^\infty \int_0^t r_t r_s (r_t - r_s) ds dt = -\frac{1}{24},$$

whereas

$$[P, Q] = \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix}.$$

Substitution in (2) gives the claimed matrix. □

6 Convex Lipschitz smoothing

The polynomial F_0 is not globally Lipschitz, so we transfer the nonzero coefficient to a Lipschitz convex model.

For $R > 1$, define the even convex C^1 function

$$\tilde{\psi}_R(u) = \begin{cases} u^4, & |u| \leq R, \\ 4R^3|u| - 3R^4, & |u| \geq R. \end{cases}$$

Convolve it with a nonnegative even C^∞ mollifier of radius $\delta_R \downarrow 0$ and call the result ψ_R . Then ψ_R is smooth, convex, globally Lipschitz, and has bounded derivatives of every positive order. Define

$$F_R(x) = \psi_R(a \cdot x) + \psi_R(b \cdot x).$$

Lemma 4. *The mollification radii may be chosen so that*

$$\mathcal{C}_{F_R}(x_0) \longrightarrow \mathcal{C}_{F_0}(x_0) \quad (R \rightarrow \infty).$$

In particular, $\mathcal{C}_{F_R}(x_0) \neq 0$ for all sufficiently large R .

Proof. Choose $\delta_R \rightarrow 0$ slowly enough that $\psi_R''(u) \rightarrow 12u^2$ at every fixed u . The piecewise function has second derivative $12u^2$ inside $(-R, R)$ and zero outside, apart from its corner in the distributional description. Its mollifications can be chosen with a uniform quadratic envelope

$$0 \leq \psi_R''(u) \leq C(1 + u^2).$$

For a twice differentiable function of at most polynomial growth,

$$D^2 P_t G(x) = e^{-2t} P_t(D^2 G)(x).$$

Gaussian dominated convergence hence gives

$$D^2 P_t F_R(x_0) \longrightarrow D^2 P_t F_0(x_0) \quad \text{for every } t \geq 0,$$

and the quadratic envelope gives the uniform bound

$$\|D^2 P_t F_R(x_0)\| \leq C e^{-2t}.$$

Thus the norm of the commutator integrand is bounded by $C e^{-2t} e^{-2s}$ on $0 < s < t < \infty$. Dominated convergence in (2), together with Lemma 3, proves the result. $\square$

7 Counterexample theorem

Theorem 5. *There exists a smooth convex globally Lipschitz function $F : \mathbb{R}^2 \rightarrow \mathbb{R}$ and $\varepsilon_0 > 0$ such that, for every $0 < \varepsilon < \varepsilon_0$, the Kim–Milman map transporting γ_2 to*

$$d\nu_\varepsilon = Z_\varepsilon^{-1} e^{-\varepsilon F} d\gamma_2$$

does not coincide with the Brenier map.

Proof. Choose R as in Lemma 4 and set $F = F_R$. By Lemma 2,

$$DS_\infty^\varepsilon(x_0) - DS_\infty^\varepsilon(x_0)^T = \varepsilon^2 \mathcal{C}_F(x_0) + o(\varepsilon^2).$$

Since $\mathcal{C}_F(x_0) \neq 0$, the derivative $DS_\infty^\varepsilon(x_0)$ is nonsymmetric for all sufficiently small positive ε .

Suppose that the forward Kim–Milman map were the Brenier map. The maps here are smooth diffeomorphisms: this follows from the smooth globally Lipschitz potential and the integrable bounds in Lemma 1. The inverse of a Brenier diffeomorphism $\nabla\phi$ is $\nabla\phi^*$, the gradient of the convex conjugate. Its derivative is symmetric wherever defined. This contradicts the nonsymmetry of $DS_\infty^\varepsilon(x_0)$. Hence the two forward maps are different. $\square$

8 Verification, limitations, and novelty

The exact algebra in Lemma 3 is independently checked by `code/verify_commutator.py` using symbolic rational arithmetic. The script is supplementary: the smoothing and perturbative ODE arguments are analytic and do not depend on numerical evidence.

The theorem is existential: it proves that every sufficiently late smooth linear-tail truncation supplies a counterexample for all sufficiently small ε , but it does not provide optimized numerical values of R , δ_R , or ε_0 . It answers map equality only; both maps remain valid contractions transporting the same measures.

The bounded novelty search on 22 July 2026 covered the run’s cheap indexes and arXiv/web searches for the exact ids 1002.0373 and 1709.06464 and the phrases “Kim–Milman map standard Gaussian Brenier”, “heat flow interpolation Brenier standard Gaussian”, and close variants. It found the source note and later applications of the Kim–Milman construction, but no rigorous later resolution of the standard-Gaussian/non-Gaussian question. This is not an exhaustive guarantee; novelty confidence is medium.

Human-review recommendation. First inspect the uniform-in-time parameter differentiation in Lemma 1; then inspect the common quadratic envelope and dominated-convergence passage in Lemma 4. The exact commutator computation has low verification risk.

References

- [1] Y.-H. Kim and E. Milman, *A Generalization of Caffarelli’s Contraction Theorem via (reverse) Heat Flow*, arXiv:1002.0373; Math. Ann. 354 (2012), 827–862.
- [2] A. Tanana, *Comparison of transport map generated by heat flow interpolation and the optimal transport Brenier map*, arXiv:1709.06464 (2017).